University of Asia Pacific

Department of Computer Science and Engineering

CSE 438: Robotics Lab

Project Proposal

ProjectTitle: **Ultra High-Speed Autonomous Line-Following Robot Precision 6-Channel IR Sensor Array and Advanced PID Control System for Maximum Accuracy and Performance**

Submitted by:

Team Name : BOD_Paradox

Section : E1

Submitted to:

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1. Title and Team Information

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2. Introduction

Line-following robots are essential mobile robots used in industrial automation, warehouses, and educational competitions for autonomous navigation along predefined paths without human intervention. The problem this project addresses is the need for a fast, accurate, and cost-effective line-following solution that can handle sharp turns and varying speeds while maintaining stability — a common challenge in real-world applications where basic 2-sensor robots overshoot or lose the line at high speeds.

This project develops a compact, DIY high-speed line-following robot using a custom 5-channel analog IR sensor array and PID (Proportional-Integral-Derivative) control algorithm. Unlike simple threshold-based followers, this robot calculates precise line position error from

multiple sensors for smooth, high-performance tracking. The design uses a veroboard chassis and off-the-shelf components, making it accessible for students and small-scale automation in Bangladesh.

3. Proposed Methodology

The project will be executed in the following sequential steps:

1. **Research & Design Phase** — Review PID control theory, differential drive kinematics, and IR sensor principles. Design the 15 cm × 9 cm veroboard chassis layout and sensor array placement.
2. **Hardware Fabrication** — Build the custom 6-channel IR sensor array using TCRT5000 sensors on perfboard/PCB. Assemble chassis, mount 600 RPM DC gear motors with wheels, L298N driver, Arduino Nano, 7805 regulator, and Li-ion battery pack.
3. **Electronics Integration & Wiring** — Connect sensors to Arduino analog pins, motor driver PWM pins, power distribution with T-connector, and add indicator LED + push button.
4. **Software Development** — Write Arduino sketch for: (a) reading 5 analog sensor values, (b) calculating weighted line position error, (c) implementing full PID algorithm, and (d) generating PWM signals for differential motor control.
5. **Calibration & Testing** — Calibrate sensors on black/white surfaces, tune PID constants (Kp, Ki, Kd) for maximum speed without oscillation, and test on straight lines, curves, and junctions.
6. **Optimization & Documentation** — Add start/stop functionality via push button, perform stress tests, document wiring diagrams, code, and performance metrics.

4. Social Impact and Motivation

This project is important because it demonstrates practical robotics for automation in developing countries like Bangladesh, where labor-intensive tasks in factories, warehouses, and agriculture can be made safer and more efficient. It benefits students (by providing affordable STEM learning tools), small industries (low-cost navigation solutions), and educational institutions (as a demonstration platform for control systems).

Motivation stems from the growing demand for skilled robotics engineers in Industry 4.0. By building everything from scratch — including the custom sensor array and rechargeable battery pack — the project promotes innovation, sustainability (rechargeable Li-ion), and self-reliance in DIY robotics, inspiring future research in autonomous systems.

5.Budget

All prices are approximate in BDT as of April 2026, sourced from local vendors including RoboticsBD, Daraz BD, and ElectroXBD.

Category	Items Included	Estimated Cost (BDT)
Microcontroller & Driver	Arduino Nano + L298N motor driver	600
Motors & Mechanics	2 × 600 RPM DC gear motors + wheels + mounts	1,100
Sensors & Array	6 × TCRT5000 IR sensors + resistors + perfboard	700
Power System	2 × 18650 Li-ion cells + T-connector + charger	500
Chassis & Misc. Electronics	Veroboard (15×9 cm), 7805 regulator, wires, LED, button, screws/nuts/standoffs, female headers	500
Total		3,400

5. Gantt Chart

Project timeline: March 2026 – May 2026

Task / Activity	Mar 2026	Early Apr	Mid Apr	Late Apr	Early May	Mid May
Project planning, procurement & design						
Fabricate 5-ch IR sensor array & chassis						
Electronics integration & wiring						
PID code development & initial testing						
PID tuning, track testing & optimization						
Documentation, final demo & presentation						

6. CEP Mapping

How Ks are addressed through the Project

Ks	Attributes	How Ks are Addressed	CLOs	PLOs
K3	Systematic, theory-based formulation of engineering fundamentals	The project applies classical control theory through PID controller formulation (error = desired - actual line position; output derived from proportional, integral, and derivative terms) for precise robot navigation.	CLO3	PLO3
K4	Engineering specialist knowledge for accepted practice areas	Specialist knowledge in embedded systems, multi-sensor data fusion, and real-time motor control using Arduino and L298N driver, representing current practice in mobile robotics.	CLO1	PLO1

K5	Knowledge that supports engineering design in a practice area	Full mechanical-electrical design of chassis, sensor array placement, and power system to support reliable line-following in industrial automation contexts.	CLO2	PLO5
K6	Knowledge of engineering practice (technology)	Practical implementation of soldering, wiring, sensor calibration, and hardware-software integration using real components and tools.	CLO4	PLO2

How Ps are addressed through the Project

Ps	Attribute	How Ps are Addressed	CLOs	PLOs
P1	Depth of knowledge required	Requires deep understanding of PID tuning, analog sensor signal processing, and real-time embedded programming to achieve high-speed stable performance.	CLO3	PLO3
P2	Range of conflicting requirements	Balances conflicting goals: high speed (600 RPM motors) vs. tracking accuracy, battery life vs.	CLO4	PLO2
		power consumption, and low cost vs. robust DIY construction.		
P7	**Interdependence**	**Hardware (sensor accuracy, motor response) is directly interdependent with software (PID parameters); changes in one subsystem affect overall system stability and performance.**	**CLO5**	**PLO5**

How As are addressed through the Project

As	Attributes	How As are Addressed
A1	Range of resources	Involves diverse resources: electronic components (sensors, motors, microcontroller), tools (soldering iron, multimeter), software (Arduino IDE), and physical testing track.
A2	Level of interaction	Requires interaction among subsystems (sensors ↔ Arduino ↔ motors), team/instructor feedback loops, and integration of mechanical, electrical, and software elements.
A5	Familiarity	Combines familiar Arduino programming with novel custom 6-channel sensor fabrication and high-speed PID optimization not covered in standard mini-projects.

7. Conclusion

This proposal presents a complete, novel, and practical line-following robot that demonstrates core robotics principles through hands-on hardware and software development. By integrating custom sensor design, PID control, and real-world testing, the project directly aligns with the course learning outcomes and CEP requirements.

Successful completion will result in a high-performance, low-cost robot ready for demonstration and future extensions (e.g., maze solving or ROS simulation integration). We look forward to presenting the working prototype by the end of the semester.

Survey Images

While attending an workshop on Line Following Robot taken by our Computer Science department's Robotics Club, we did a survey on our project **Ultra High-Speed Autonomous Line-Following Robot Precision 6-Channel IR Sensor Array and Advanced PID Control System for Maximum Accuracy and Performance.**

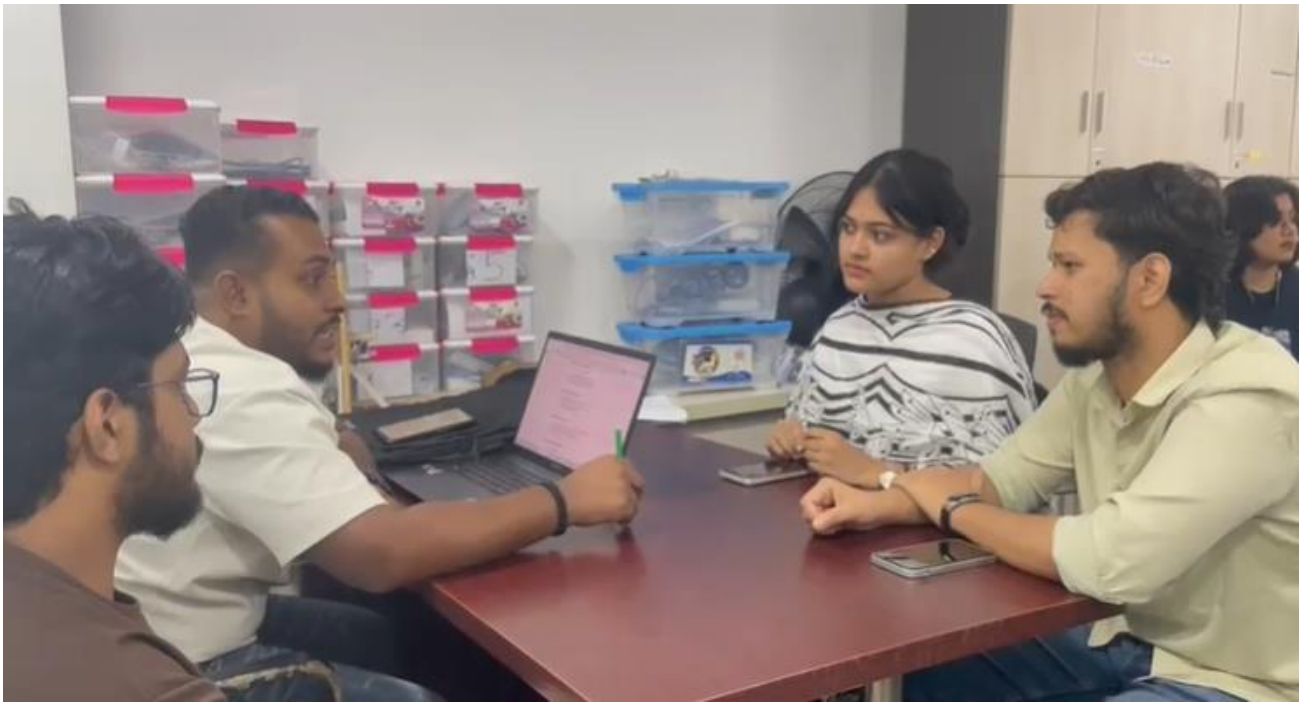

Figure 1: Taking Survey about hardware in the Project

Figure 2: Attending Workshop